Capstone: Problem Definition

Refined Problem Statement, Stakeholder Analysis, & SMART Criteria

Due: Week 10 | Points: 30

Instructions

This milestone builds on your approved proposal from Week 8. Your goal is to refine and deepen the problem definition—not just restate what you already wrote.

By the end of this deliverable, a reader should understand exactly what problem you're solving, for whom, how you'll measure success, and what has changed since your initial proposal. Submit this completed template alongside your Week 10 milestone check-in.

Project Title:

Revolutionizing Pricing Practices in a Small Christian Business

Business Partner: Angelyn Kornelsen **CS Partner:** Lesther Ouma

Section 1: Refined Problem Statement (Business Partner Lead)

Rewrite your problem statement from the proposal. It should now be more specific and grounded in what you've learned since Week 8.

Problem statement (3–5 sentences):

Kornelsen Solutions Inc. needs a standardized job pricing system because pricing is currently done from memory and intuition with no documented formula. Currently, jobs are quoted inconsistently with variable prices for similar work and relationship-based adjustments that have no defined baseline which causes unpredictable revenue, wasted time and mental energy on every quote, and no visibility into exact profit margins per job. With data analysis, we can build a pricing formula that accounts for door type, size, materials, labor hours, travel distance, and customer tier which will reduce quoting time and effort to near zero, enforce price consistency across all jobs, and make profitability measurable on every single transaction.

What Changed from the Proposal?

Describe how your understanding of the problem has evolved since Week 8. What did you learn, refine, or narrow? If nothing changed, explain why the original framing was already sufficient.

- We had a clear and concise problem statement, with stakeholders well identified and the scope well defined. Additionally, one team member was familiar with the dataset, giving us a valuable perspective we would not have otherwise had.

Section 2: Stakeholder Analysis (Business Partner Lead)

Identify the specific person or role who will use your analysis. “The business” or “management” is not specific enough.

	Your Response
Primary Decision-Maker	Royden Kornelsen
Their Role / Title	Business Owner
Organization	Kornelsen Solutions Inc
What decisions do they make?	All business decisions: From general and big to the small ones
What information do they currently lack?	Royden lacks a simple, reproducible, consistent pricing system
How will they use your deliverable?	To quote invoices and come up with price estimates

Other Affected Stakeholders

Who else is affected by this problem or will be impacted by your recommendations? List 2–3 additional stakeholders and briefly note their interest.

Stakeholder	Interest / Concern
Employee	Maybe in charge of implementing the system or use it in the absence of business owner
Recurring Customers	Pricing system will affect all customers and may not be able to adjust to loyalty etc.

Section 3: SMART Success Criteria

Define 3–4 success criteria using the SMART framework. Each criterion should be a concrete, verifiable outcome—not a vague aspiration.

SMART Element	Meaning
Specific	What exactly will be produced or answered?
Measurable	How will you know it was achieved?
Achievable	Is it realistic given your data, skills, and timeline?
Relevant	Does it matter to your stakeholder?
Time-bound	When will it be complete?

Criterion 1:

Within 4 weeks of implementation, the program will generate quotes that match the business owner's pricing decisions with at least **90% accuracy**, measured by comparing 100 sampled quotes against the owner's baseline.

Criterion 2:

Within 4 weeks, pricing variability across similar jobs will decrease by **30%**, measured by analyzing standard deviation in quotes for comparable inputs before and after implementation.

Criterion 3:

At least **90% of new users** will successfully complete a full pricing task without assistance, measured through usability testing sessions on adoption.

Section 4: Data & Methodology Preview (CS Partner Lead)
Provide an early look at your technical approach. This does not need to be final—it will be refined in the Data Acquisition milestone (Week 11).

Data source(s) identified:

The Business' CRM system

Current data status: X Secured ☐ Access requested ☐ Partially obtained ☐ Still identifying

Preliminary analytical approach (What methods or tools do you expect to use? Why?):

- Pandas for data analysis
- SQL for data mapping and
- Excel for data cleaning

Known data risks or gaps:

- We may have inconsistent data entries (one type of data being entered in different fields) due to human error or a lapse in judgement.
- We may have irrelevant data (too old to make informed decisions on current pricing) due to inflation and/or other organizational changes.
- Because of these data risks, our final product may be less predictively useful than originally hoped for or planned.

Section 5: Progress Since Proposal

Business Partner's Contributions	CS Partner's Contributions
The business partner has provided access to the data and has insider knowledge on the business and the way it runs. She also has personal connection to the owner and she has been in contact with him about the proposal.	The CS partner contributed to the SMART goals and the success criteria required. He also helped to identify the key risks and gaps as well as gave the preliminary analytical approach to creating the solution to the problem.

Key decisions made since the proposal:

- Tools we are going to use

Next steps for the Data Acquisition milestone (Week 11):

1. Download the data from QuickBooks CRM
2. Remove personally identifying information (PII)
3. Make sure all needed fields are filled in accurately

Submission Checklist

- ☐ Refined problem statement (not just a copy of the proposal)
- ☐ Primary stakeholder identified with specific decision-making context
- ☐ Additional affected stakeholders listed
- ☐ 3–4 SMART success criteria
- ☐ Data and methodology preview
- ☐ Progress evidence and partner contributions documented
- ☐ Week 10 milestone check-in form (separate document)